

Appendix X — Notation Index, Appendix Y — List of Acronyms, Bibliographic Notes



7月 19, 2026

Appendix X

Notation Index

This appendix summarizes the principal mathematical symbols, operators, and abbreviations employed throughout the *Social Quantum Field Theory* (SQFT) Research Series.

A. Mathematical Symbols

Symbol	Meaning
F	Social field
H	Hilbert space
ρ	Density operator
H	Hamiltonian
L_k	Lindblad operator
$ \psi\rangle\langle\psi $	Pure-state projector
$\hat{A}$	Observable
$U(t)$	Time-evolution operator
$\hbar$	Reduced Planck constant
S	Entropy
Z	Partition function
Γ	Configuration space
G	Symmetry group
∇	Covariant derivative
$\otimes$	Tensor product
$\oplus$	Direct sum
$[A, B]$	Commutator
$\{A, B\}$	Anticommutator

B. Frequently Used Equations

Schrödinger Equation

$$i\hbar\frac{\partial}{\partial t}\psi = H\psi$$

Heisenberg Equation

$$\frac{dA_H}{dt} = \frac{i}{\hbar}[H, A_H] + \frac{\partial A_H}{\partial t}$$

Lindblad Master Equation

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right)$$

Path Integral

$$\int D\phi e^{iS[\phi]/\hbar}$$

Renormalization Group Flow

$$\beta(g) = \mu \frac{dg}{d\mu}$$

Appendix Y

List of Acronyms

Acronym	Meaning
SQFT	Social Quantum Field Theory
QFT	Quantum Field Theory
QM	Quantum Mechanics
QI	Quantum Information
RG	Renormalization Group
EFT	Effective Field Theory
QZE	Quantum Zeno Effect
AI	Artificial Intelligence
ML	Machine Learning
TN	Tensor Network
MPS	Matrix Product State
PEPS	Projected Entangled Pair States
MERA	Multi-scale Entanglement Renormalization Ansatz
CFT	Conformal Field Theory
OQS	Open Quantum Systems

Acronym	Meaning
CPTP	Completely Positive Trace-Preserving
UV	Ultraviolet
IR	Infrared

Bibliographic Notes

The present volume was developed by synthesizing ideas from several mature mathematical traditions while proposing an original relational framework.

The mathematical language employed throughout this work is primarily informed by

- quantum mechanics,
- quantum field theory,
- open quantum systems,
- differential geometry,
- topology,
- tensor network methods,
- statistical physics,
- information theory,
- category theory,
- and sociological field theory.

The interpretation of **social systems as relational field structures**, together with the formal concepts introduced under the name **Social Quantum Field Theory (SQFT)**, constitutes the principal original contribution of this volume.

Readers are encouraged to consult the References for foundational literature before extending or modifying the proposed framework.

No single mathematical tradition is assumed to provide a complete description of complex relational systems. Rather, this work advocates a unified formal perspective whose validity must ultimately be judged through mathematical rigor, logical consistency, computational reproducibility, and independent scholarly evaluation.

Closing Editorial Statement

This First Edition has been prepared as a complete and self-contained research monograph.

Any future editions should preserve a clear historical record of revisions while continuing to strengthen the mathematical foundations, simplify the formalism where possible, and expand the scope of rigorous applications.

Errors, omissions, and limitations that remain in this edition are the sole responsibility of the author and should be regarded as opportunities for future refinement.

Publication Information

Title

Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields

Author

Chou I-Hsien

First Edition

2026

Recommended Citation

Chou, I.-H. (2026). *Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields*. First Edition.

Colophon

Typeset using Unicode mathematical notation and prepared as a research monograph in accordance with contemporary international academic publishing conventions.

Final Colophon

"A book ends where inquiry begins."

End of First Edition

Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields

Author: Chou I-Hsien

First Edition — 2026

Ad Veritatem Per Rationem

("Toward truth through reason.")



輸入留言

這個網誌中的熱門文章

Foundational Charter of the Social Quantum Field Theory Research Community

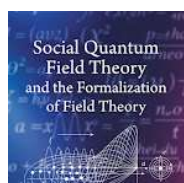
7月 19, 2026

The SQFT Charter Foundational Charter of the Social Quantum Field Theory Research Community Preamble The Social Quantum Field Theory (SQFT) Research Community is founded upon the conviction that mathematics advances through open inquiry ...

[閱讀完整內容](#)

社會量子場論（SQFT）超越布迪厄的場域理論，建構關係社會學的形式化 1~10

7月 15, 2026



預印本 社會量子場論與場論的形式化 周易先 獨立研究者 · 版本 2.0（數學版） · 2026年7月 摘要 本文提出社會量子場論（SQFT）作為用以表示複雜社會系統中關係動態的形式數學框架。SQFT並不主張社 ...

[閱讀完整內容](#)

量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響

9月 27, 2025

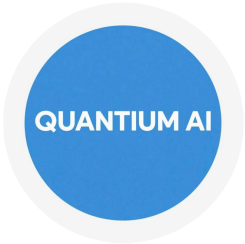


量子之影：台灣QNF-3量子導航系統的崛起與其地緣政治影響 摘要 在2035年，全球局勢緊張之際，台灣的量子導航技術成為重塑國際權力格局的關鍵力量。本篇科幻論文式敘事，記錄了國家中山科學研 ...

 技術提供：Blogger

主題圖片來源：[Michael Elkan](#)





IH

瀏覽簡介

封存

▼

檢舉濫用情形